Your Name: Solutions from Instructor Roll. No.: TA name:	Duration: 3 hours Max. marks: 150

End-sem Exam

EE698V – Machine Learning for Signal Processing

Use extra sheets for rough work. They won't be graded but should be submitted to validate your efforts.

Q. The probability of a binary random variable $x \in \{0,1\}$ is given by $p(x|\mu) = \mu^x(1-\mu)^{1-x}$

i. The mean, $E[x]$, is given by μ [2]

ii. The variance, $\mathrm{var}[x]$, is given by $\mu(1-\mu)$ [2]

iii. The entropy, $H[x]$, is given by $-\mu \log \mu - (1-\mu)\log(1-\mu)$ [2]

Q. I have got 10 flowers, which I give to two annotators to label as rose (R) and lotus (L). The first annotator gives me the 10 labels (in order) as [RRRRR RRR LL] and the second one returns [RL RL RL RL RL], respectively.

The Cohen's Kappa score for inter-annotator agreement is 0 [4]

Q. A set of orthonormal unit vectors $\boldsymbol{u}_i;\ i = 1, \ldots, N$ span a vector space; i.e., $u_i^T u_j = 1$ if $i = j$ and 0 otherwise.

i. Given a vector $\boldsymbol{x}$, what is its projection in the subspace spanned by only odd indexed $\boldsymbol{u}_i$s? [1]

$$\sum_{i=\{1,3,5,\ldots\}} (x^T u_i)\, u_i$$

ii. What is the distance of point $\boldsymbol{x}$ from the origin? [1]

$$(x^T x)^{1/2}$$

iii. What is the projection of point $\boldsymbol{x}$ on the surface defined by $\boldsymbol{w}^T\boldsymbol{z} + w_0 = 0$? (hint: remember the linear discriminant) [4]

$$x - \frac{w}{\|w\|}\left(\frac{w^T x + w_0}{\|w\|}\right)$$

Q. Given a set of vectors $\boldsymbol{x}_i;\ i = 1, \ldots, N$, we want to transform them linearly as $\boldsymbol{y}_i = f(\boldsymbol{x}_i)$ such that $\boldsymbol{y}_i$ have unit variance. What should be $f(\boldsymbol{x}_i)$? (Use something like PCA) [2]

$$f(x_i) = \Lambda^{-1/2} E\, \tilde{x}_i$$

here, $E^T \Lambda E = S$ is eigenvalue decomposition of $S = \frac{1}{N}\sum_{i=1}^{N} \tilde{x}_i \tilde{x}_i^T$, with $\tilde{x}_i = x_i - \frac{1}{N}\sum_i x_i$

Q. The no. of learnable parameters in a max-pooling layer of filter size F=3 and stride size S=2 is 0 [2]

Q. The input to a convolutional layer with 11x11 filter with stride 1 is an RGB image of size 15x15. In order to keep output size same as the input size, a padding of size P= 5 along the border can be used. [2]

Q.

```
model.add(Dense(24, input_dim= 100, activation='relu'))
model.add(Dense(2, activation='softmax'))
```

In the above python code, using a sequential model from keras library, the total number of weights to be trained (including the biases) is $101 \times 24 + 25 \times 2 = 2474$ [4]

Q. In the Fig. CNN-1,

i. the total number of learnable parameters (weights) is 1670 or 1650 (w/o bias) [4]

ii. The size of the output of the last layer is $4 \times 4 \times 10$ [4]

iii. The computational complexity (number of multiplications) in the forward pass is close to $41{,}400$ [6]

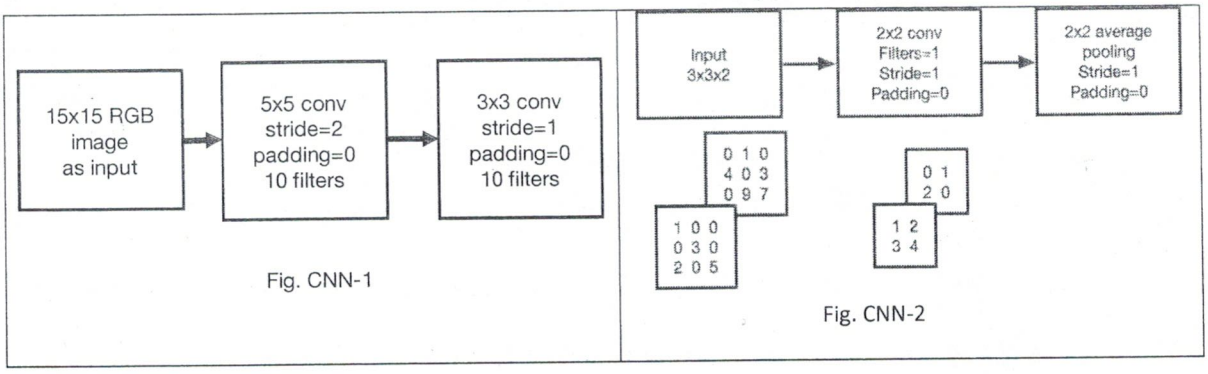

Fig. CNN-1 Fig. CNN-2

Q.[9] For the Fig. CNN-2, the values of the input and the filter are specified. The final output will be

$\frac{87}{4}$ or 21.75 [6]

Q.[10] The pdf of a random variable $x \in R^N$ is modelled using a GMM with K components

i. The total number of learnable parameters in the model is $(K-1)+KN+KN^2$ or $K+KN+KN^2$ ↑ acceptable [2]

ii. Assuming that all dimensions of each Gaussian component are uncorrelated, the total number of learnable parameters will be $(K-1)+KN+KN$ or $K+KN+KN$ ↑ acceptable [2]

Q.[11] Dynamic time warping involves optimizing: $\min_{\pi} D(\pi_{1:K})$, K is the length of the reference sequence.

i. To solve it efficiently, we use Bellman's Optimality principle, which simplifies the equation as $\min_{\pi} D(\pi_{0:k}) = \min_{\pi_k} d(\pi_k) + \min_{\pi} D(\pi_{0:k-1})$ [1+3]

ii. Given two sequences of characters JACKSON and JAIKISHAN, we need to align the two using DTW. The cost $d(i,j)$ is 0 if the two characters c_i, c_j are same and 1 otherwise. Only three kind of local transition steps are allowed in forward direction - 1 vertical, 1 horizontal and 1 diagonal. Along the best matching path, the min cost is 4. Using JACKSON as the reference, the corresponding other sequence will be JAIKSHN or JAIKSAN [6+4]

(Info: you can use it to teach correct pronunciation)

Q.[12] Bag of features based representation for time series loses the dynamic or temporal or sequence information but preserves the co-occurance information. [2+2]

Q.[13] What is the maximum number of consecutive features of a time series the following can model: [5]

i. A 2nd order Markov chain: 3
ii. HMM (1st order Markov): no limit
iii. GMM: 1
iv. RNN: no limit
v. CNN: F, i.e., length of filter along the time axis

Q.[14] The following sequential data is given: "**dhananjayavivardhanam dhanamududha govardhanam susadhanamabadhanam sumanasam samaradhanam**" on which we want to train a character-level markov model. Ignoring white-spaces in data, compute the following probabilities. [3x4]

i. P(**d**) = 9/84

ii. P(**dhana**) = $7/(84-5+1) = 7/80$ (hint: joint probability)

iii. P(**a**|**dhan**) = $7/7 = 1$ (**a** follows **dhan**)

iv. P(**m**|**dhana**) = 6/7

(Info: You can use such models as language models for speech recognition)

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Q. A sequence $x_n \in R^D$; $n = 1, \ldots, N$ is modelled using an HMM with hidden state $z \in \{1, \ldots, K\}$. The HMM parameters are estimated as $\pi(i), A_{ij}, p(x|z = i)$. Express the following in terms of HMM parameters [2x5]

i. $p(z_n = j|x_n, z_{n-1} = i) =$ $p(x_n|z_n=j)\, A_{ij} \Big/ \left[\sum_{j'} p(x_n|z_n=j')\, A_{ij'}\right]$

ii. Log likelihood of $x_1, x_2, \ldots x_N =$ $\log\left[\sum_{z_1, z_2, \ldots z_N} \pi(z_1)\left(\prod_{n=2}^{N} A_{z_{n-1}z_n}\right)\left(\prod_{n=1}^{N} p(x_n|z_n)\right)\right]$

iii. The prior, $p(z_1, \ldots, z_N) =$ $\pi(z_1)\left(\prod_{n=2}^{N} A_{z_{n-1}z_n}\right)$

iv. The posterior, $p(z_1, \ldots, z_N|x_1, \ldots, x_N) =$ $\dfrac{\pi(z_1)\left(\prod_{n=2}^{N} A_{z_{n-1}z_n}\right)\left(\prod_{n=1}^{N} p(x_n|z_n)\right)}{\sum_{z_1,\ldots,z_n} \pi(z_1)\left(\prod_{n=2}^{N} A_{z_{n-1}z_n}\right)\left(\prod_{n=1}^{N} p(x_n|z_n)\right)}$

v. $p(x_1, \ldots, x_N|z_1, \ldots, z_N) =$ $\prod_{n=1}^{N} p(x_n|z_n)$

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Q. Coin flipping experiment: Ram has two coins - C_1, C_2 - one with $P(H) = 0.75$ and the other with $P(H) = 0.25$, respectively. He randomly (uniformly) picks one coin at $n = 1$ and starts flipping to observe $x_n \in \{H, T\}$, while switching (or changing) the coin with a probability 0.25. He observes the sequence "HHTH HTTT".

i. The most likely sequence of coins is $C_1C_1C_1C_1$ $C_2C_2C_2C_2$ [8]
(hint: Viterbi decoding)

ii. If the sequence of coins were "$C_2C_2C_2C_2\ C_1C_1C_1C_1$", the probability of observing the above sequence would be $\left(\frac{1}{4}\times\frac{1}{4}\times\frac{3}{4}\times\frac{1}{4}\right)\times\left(\frac{3}{4}\times\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}\right) = \frac{3^2}{4^8}$ [2]
(leave your answer as powers of terms, no need to calculate the final value)

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Q. Out of template matching, HMMs, CNNs and RNNs: [2]

i. The one which benefits the most from GPUs (parallel computing) is CNNs

ii. The one which is fastest to deploy with less data is template matching

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Q. In Fig. 2, we want to compress the data to 1D [2]

i. If we use PCA, draw (on Fig 2) the line along which the data can be best projected.

ii. If we want to attain good classification accuracy (star vs circle) in the 1D space, the compression method we should use is Fischer's discriminant method

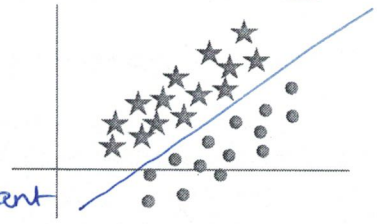

Fig. 2

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Q. The rms voltage in a room's switchboard was measured many times over a month and was found to have the values [241.0, 239.1, 241.1, 239.6, 242.1, 240.1, 238.4]

i. If we model it with a Laplace distribution, ML estimate of its $\mu =$ 240.1 (median) and $\beta =$ 1.014 [2]

ii. If we model it with a uniform distribution, ML estimate of its $a =$ 238.4 (min) and $b =$ 242.1 (max) [2]

iii. If we model it with a Gaussian distribution, ML estimate of its $\mu =$ 240.2 and $\sigma =$ 1.19 [2]

iv. To find which of the above three pdf's models it the best, we will compare: log likelihood [1]

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Q. Given the data samples $\boldsymbol{s}_n; n = 1, \ldots, N$, from two classes C_1, C_2. We want to project them to one dimension using $y_n = \boldsymbol{w}^T\boldsymbol{s}_n$. To find the optimal $\boldsymbol{w}$, the cost function is defined as $J(\boldsymbol{w}) = \frac{(m_2-m_1)^2}{s_1^2+s_2^2}$, where m_i are the means of y_n from C_i. We can write $J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}$

i. Here, $S_B =$ $(\mathbb{m}_2 - \mathbb{m}_1)(\mathbb{m}_2 - \mathbb{m}_1)^T$; $\mathbb{m}_i = \sum_{n\in C_i} \mathbb{s}_n / N_i$; $i = 1, 2$ [2]

ii. And, $S_W =$ $\sum_{n\in C_1} (\mathbb{x}_n - \mathbb{m}_1)(\mathbb{x}_n - \mathbb{m}_1)^T + \sum_{n\in C_2} (\mathbb{x}_n - \mathbb{m}_2)(\mathbb{x}_n - \mathbb{m}_2)^T$ [2]

iii. Derive the optimal value of $\boldsymbol{w}$. The final expression is $\boldsymbol{w}$ = $S_W^{-1}(m_2 - m_1)$ [2]

(write derivation below) [4]

$$\frac{\partial J}{\partial w} = 0 \Rightarrow \underbrace{(w^T S_B w)}_{\in \mathbb{R}} S_W w - \underbrace{(w^T S_W w)}_{\in \mathbb{R}} S_B w = 0$$

$$S_W w = \underbrace{(m_2 - m_1)}_{S_B} \underbrace{(m_2 - m_1)^T w}_{\text{scalar}} \times \text{scalar}$$

$$\therefore \boxed{w = S_W^{-1}(m_2 - m_1)} \times \text{scalar}$$

we want only direction so scalar can be omitted

21 **Q.** In the NMF derivation, $F(\boldsymbol{h}) = \frac{1}{2}\sum_i (v_i - \sum_a W_{ia} h_a)$. The Taylor's expansion of $F(\boldsymbol{h})$ about $\boldsymbol{h}^t$ is given by

$$F(h^t) + \sum_{a'} (h - h^t)_{a'} \left(\sum_i \left(v_i - \sum_a w_{ia} h_a\right)(-w_{ia'})\right) + \frac{1}{2} \sum_{aa'} (h - h^t)_a (W^T W)_{aa'} (h - h^t)_{a'}$$ [5]

22 **Q.** Gopal designed a new classifier which he claimed works better than Mohan's. We can do a sign test to find out if the two are really different (statistically). So we performed several independent tests on which the accuracies of the two were found to be

	Test 1	Test 2	Test 3	Test 4	Test 5	Test 6	Test 7	Test 8
Gopal's	87.4%	88.3%	81.0%	79.1%	77.4%	90.3%	58.7%	76.5%
Mohan's	90.9%	86.1%	80.9%	79.2%	76.8%	86.7%	54.3%	75.2%

i. The null hypothesis is: both the algorithms are 'same'. [1]

ii. The alternate hypothesis is: both the algorithms are 'different'. [1]

iii. Assuming uniform binomial distribution, the p-value = $\frac{37}{2^7} = \frac{37}{128} = 0.29$ [6]

iv. The result of this sign test is: null hypothesis is 'accepted'. [1]

Multiple choice questions (one correct; mark ✓):

23 **Q.** You have 10K unlabelled samples for rose vs lotus classification. You randomly picked 1K samples, got them annotated and trained a classifier to predict P(rose|sample). Now, you have to pick another batch of 1K unlabelled samples to give for annotation. Which one would help the model improve the most? [2]

- ☐ The ones with P(rose|sample) close to 0
- ☐ The ones with P(rose|sample) close to 1
- ☑ The ones with P(rose|sample) close to 0.5
- ☐ Pick randomly

24 **Q.** For hyperparameter tuning, which of the following is found to work the best for large search spaces [2]

- ☐ Grid search with narrow grid
- ☐ Grid search with wide grid
- ☐ Gradient descent with adaptive step size
- ☑ Pick randomly

25 **Q.** The effective receptive field of a stack of L layers of FxF convolution filters, with stride S (< F) at each layer, is same as that of one MxM convolutional layer. Here, M is given by [4]

- ☑ $M = FS^{L-1} + (F - S)(1 + S + \cdots + S^{L-2})$
- ☐ $M = FS^L + (F - 1)(1 + S + \cdots + S^{L-2})$
- ☐ $M = FS^{L-1} + (F - S)(L - 1)$
- ☐ $M = FS^L + (F - 1)(L - 1)$
- ☐ None of the above

Multiple choice questions (mark ✓ for true and ✗ for false; more than one can be correct; 0.5 mark for TP and TN; 0 for FP and FN):

26 **Q.** Choose the true statement(s): [2]

- ☑ CNN layers are computationally more expensive than fully connected layers while prediction (forward pass)
- ☒ CNN layers are computationally less expensive than fully connected layers while prediction (forward pass)
- ☒ CNN layers generally have more number of parameters as compared to fully connected layers
- ☑ CNN layers generally have less number of parameters as compared to fully connected layers